

Chapter 4: User Requirements Gathering and Analysis - 2



Methods of Gathering User Requirements

- User survey
- User interview
- User existing documents review
- User existing software review
- Study user-supplied documentation

User Survey

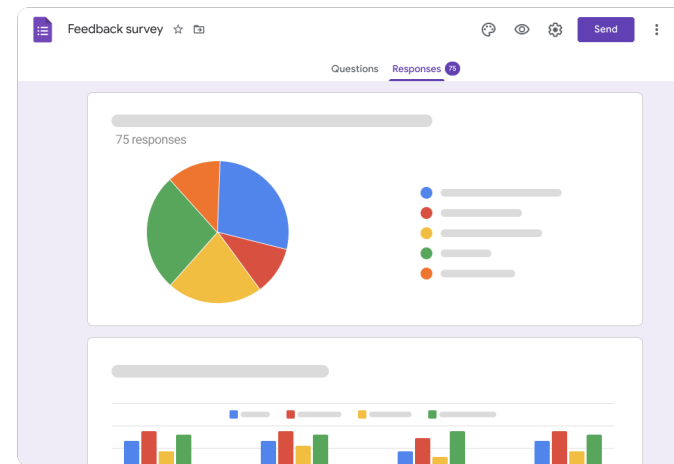
Form of questioning.

Advantages:

- Can be conducted easily
- Can reach many people with low cost

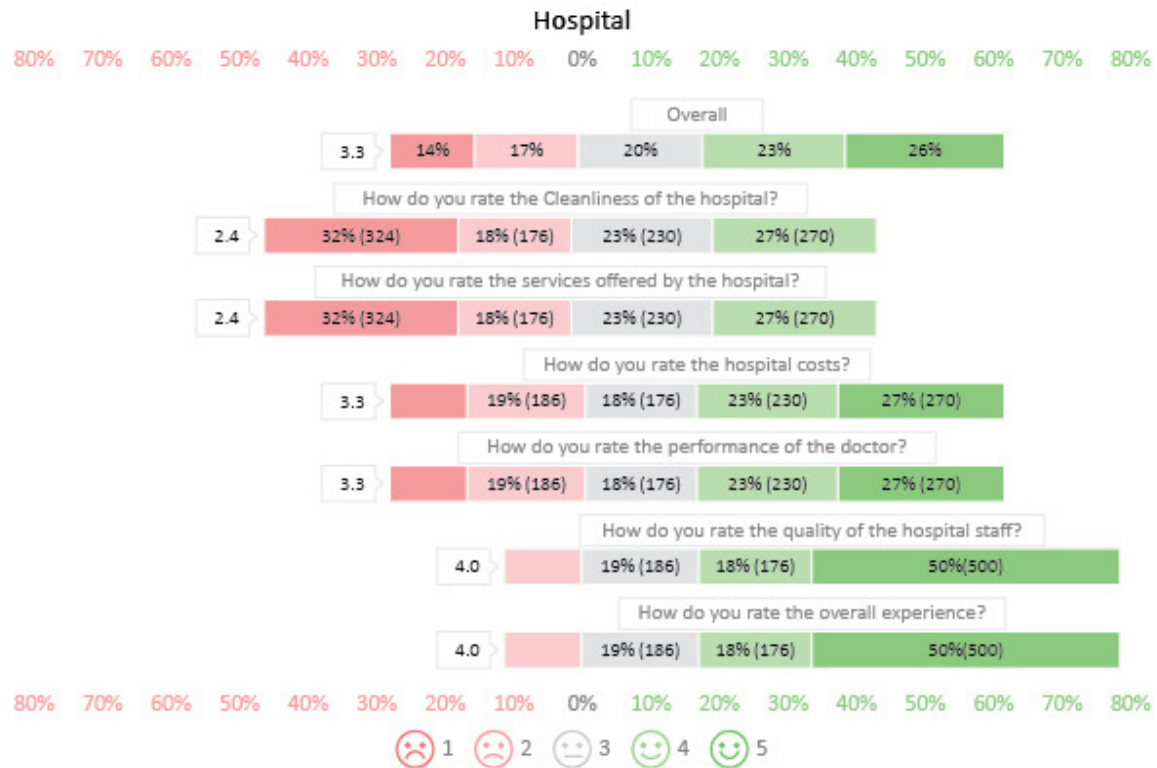
Disadvantages:

- Responses may not be useful



Visualise Survey Data

Questions	Rating	Response Count
How do you rate the cleanliness of the hospital?	1	324
How do you rate the cleanliness of the hospital?	2	176
How do you rate the cleanliness of the hospital?	3	230
How do you rate the cleanliness of the hospital?	4	278
How do you rate the cleanliness of the hospital?	5	0
How do you rate the performance of the doctors?	1	138
How do you rate the performance of the doctors?	2	186
How do you rate the performance of the doctors?	3	176
How do you rate the performance of the doctors?	4	230
How do you rate the performance of the doctors?	5	270
How do you rate your overall experience?	1	0
How do you rate your overall experience?	2	138
How do you rate your overall experience?	3	186
How do you rate your overall experience?	4	176
How do you rate your overall experience?	5	500
How do you rate the quality of the hospital staff?	1	0
How do you rate the quality of the hospital staff?	2	138
How do you rate the quality of the hospital staff?	3	186
How do you rate the quality of the hospital staff?	4	176
How do you rate the quality of the hospital staff?	5	500



User Interview

Can be conducted

- one-on-one
- small group

Advantages:

- Interviewer can guide interviewee during the interview
- Encourages contact between developers and users

Disadvantages:

- Time consuming



Existing Documents Review

Data reviewed can be

- Hardcopy or
- Electronic copy / softcopy

May include

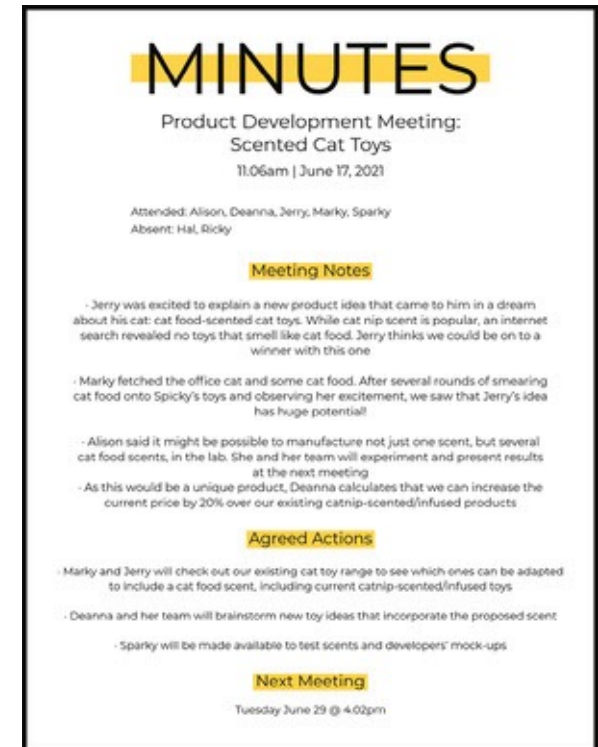
- reports,
- program logs,
- meeting minutes,
- newsletters, and
- marketing materials

Advantages:

- Relatively inexpensive
- Good source of background information

Disadvantages:

- Information may be incomplete or inaccurate



Existing Software Review

Verification of both functional and non-functional requirements

Advantages:

- Identify design defects
- Highlights areas of existing system for improvement

Disadvantages:

- Time consuming to review the code
- Increased project implementation cost if the developers' time has to be paid to review



Study User-supplied Documentation

Advantages:

- No time commitment from users is required
- Good for learning about
 - procedures,
 - regulations, and
 - standards

Disadvantages:

- Day-to-day work may differ from documented procedures

Software Requirements Document (SRD)

An official document

- of what is required of the system developers.
- specifies WHAT the system should do rather than HOW it should do.

Includes both

- user requirements, and
- system requirements

It is NOT a design document.

User Requirements Document (URD)

User Requirements

- also known as User Needs.
- describe what the user does with the system.
 - such as what activities users must be able to perform.

URD is a document used in software engineering specifies what the user expects the software to be able to do.

Components in a URD

Often URD includes priority ranking for each requirement.

A typical URD may include the followings:

1. Mandatory requirements
2. Desirable requirements
3. Optional requirements
4. Possible future enhancements

Mandatory Requirements

- Highest priority.
- This feature must be built into the final system.
- Compulsory requirement that has to be part of the bid offer.
- Failure to meet the requirement means that the bid fails the requirement and will be eliminated from the tender process.

Desirable Requirements

- This feature should be built into the final system unless the cost is too high.
- It means the requirement is important but not mandatory.

Optional Requirements

- This feature can be built into the system.
- Something that is in scope, but not necessarily required by the system.

Possible Future Enhancements

- This feature may be used in the final system.
- Budget and project timeline are key factors leading to schedule for future development of new features in the system.

Requirements Checklist

A tool to assist in determining whether the requirements document is

- correct,
- complete,
- unambiguous,
- consistent,
- verifiable and
- approved.

Example:

- Check common language: Make sure every acronym can be found in the glossary or data dictionary.

Summary

1. Methods of gathering user requirements.

- User survey,
 - Can be conducted easily
 - Can reach many people with low cost
- User interview,
 - Interviewer can guide interviewee during the interview
 - Encourages contact between developers and users
- User existing documents review,
 - Data can be hardcopy or softcopy
- User existing software review,
 - identify design defects
 - highlight areas of existing system improvement
- Study user-supplied documentation
 - Good for learning about procedures, regulations, and standards
 - No time commitment from user required

Summary

2. Software Requirement Document (SRD) is an official document

- of what is required of the system developers.
- specifies WHAT the system should do rather than HOW it should do.
- Includes both
 - user requirements
 1. mandatory requirements - feature must be built into the final system.
 2. desirable requirements - feature should be built into the final system unless the cost is too high.
 3. optional requirements
 4. possible future enhancement
 - system requirements

Summary

3. Requirements Checklist is a tool to assist in determining whether the requirements are documented correctly and completely.

Example:

- Check common language: Make sure every acronym can be found in the glossary or data dictionary.

References

1. Complete Guide To Requirements Gathering Process + Template,
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2. Product Requirements Document,
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